

VARLEE SALIA-MAKULAKO FOFANA

Full-Stack Software Engineer

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PROFESSIONAL EXPERIENCE

Software Engineer

myScholy | Remote | February 2025 to Present

- Built and operate a live scholarship platform end to end: React and Vite front end, Django REST Framework API, PostgreSQL, Redis and a Gemini-powered assistant, running on Oracle Cloud infrastructure I provisioned with a warm standby on Render sharing one database.
- Engineered JWT authentication with role-based access control, plus a version-scoped Redis cache where a single write invalidates every derived key, eliminating stale reads without enumerating filter combinations.
- Every push runs 200+ unit tests and Playwright end-to-end tests before deploying. A scheduled digest emails students 5 live scholarships every 72 hours, guarded against double-sends by a database record.

Freelance Full-Stack Web Developer

Freelance | Remote | 2024 to Present

- Sole developer on client web applications of five to six public routes each, owning the PostgreSQL schema, REST API, React front end and authenticated admin from first requirement through to production deployment.
- Built CMS-style admin dashboards spanning six screens for content, people, programs and enquiries, giving clients full control of their own content without developer involvement.
- Standardized delivery on a reusable React 19, Vite and Tailwind CSS component library with Zod-validated forms and accessible Radix UI primitives, cutting setup time on every new build. All client sites remain in continuous service.

Research and Data Analyst Intern

Educate Children | Monrovia, Liberia | June 2023 to December 2023

- Collected and transformed field data from structured interviews into validated Excel datasets, improving data capture accuracy by 55% through standardized entry rules and systematic treatment of missing values.
- Reduced recurring data processing time by two hours per cycle, replacing manual cleaning with repeatable transformation and validation workflows.
- Built pivot table and chart dashboards that surfaced program indicators for non-technical staff, feeding program evaluation and strategic planning across the organization's education initiatives.

PROJECTS

ChatTobira: AI Study Assistant (RAG) | Software Engineer | 2026

Next.js, TypeScript, Cloudflare Workers, PostgreSQL, pgvector, Python, Gemini API

- Built a retrieval-augmented study assistant serving classmates on the same university course, answering Japanese-language questions from the course corpus with citations to the printed textbook page, and generating practice papers from a format specification tabulated across 40 real past exams.
- The source PDFs carried no usable text layer, a 216-page textbook yielding 3 extractable characters, so ingestion renders and transcribes every page through a vision model, checkpointed per page so rate-limited runs resume without reprocessing.
- Implemented hybrid Japanese retrieval fusing morphological segmentation with pgvector similarity by reciprocal rank fusion, and held running cost near \$0/month through a semantic answer cache and three-provider failover.

Revelle Beauty: E-Commerce Storefront | Software Engineer | 2026

React 19, TypeScript, Express 5, PostgreSQL, Stripe, Docker, Cloudflare Workers

- Built a full commerce pipeline covering catalog, server-side cart, Stripe checkout, order tracking and admin console across 42 live product shades. The browser never states a price; a single pricing service reprices every order from the database.
- Reduced each catalog endpoint to one SQL statement using JSON aggregation and a window function, eliminating N+1 reads across the entire catalog read path.
- Hardened the trust boundary with deny-all row-level security, bcrypt-to-JWT admin auth in an httpOnly cookie and raw-body Stripe webhook verification, covered by 92 automated tests across Vitest and Playwright.

EDUCATION

Bachelor of Business Administration, Accounting & Finance | Ritsumeikan Asia Pacific University | April 2025 to Present

High School Diploma | African Leadership Academy, Johannesburg, South Africa | August 2020 to June 2022